Pentagonal Number Theorem

JC PM MV

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1 Partitions

Definition 1. A *partition* of a positive integer n, also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n.

Example 2. $5 = 4+1 = 3+2 = 3+1+1 = 2+2+1 = 2+1+1+1 = 1+1+1+1+1$
$p(5) = 7$

Lemma 3. *The generating function for* $p(n)$ *is given by*

$$\sum_{n=0}^{\infty} p(n)\, x^n = \prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \ldots) = \prod_{k=1}^{\infty} (1 - x^k)^{-1}.$$

Definition 4. For a positive integer n we denote by $p_e(n)$ the number of partitions of n into an even number of *different* parts, analogously, we denote by $p_o(n)$ the number of partitions of n into an odd number of *different* parts.

Lemma 5. *We have*

$$\prod_{k=1}^{\infty} (1 - x^k) \;=\; \sum_{s=0}^{\infty} \sum_{k_1 < k_2 < \cdots < k_s} (-1)^s\, x^{k_1 + k_2 + \ldots + k_s} \;=\; \sum_{n=0}^{\infty} c_n\, x^n$$

where $c_0 = 1$ *and for* $n \geq 1$ *and* $c_n = p_e(n) - p_o(n)$.

Example 6. *We compute* $\prod_{i=1}^{20} (1 - x^i) + o(x^{20})$.

The answer is $1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + O\left(x^{20}\right)$.

n		$p_o(n)$	$p_e(n)$
1	1	1	0
2	2	1	0
3	$3 = 2 + 1$	1	1
4	$4 = 3 + 1$	1	1
5	$5 = 4 + 1 = 3 + 2$	1	2
6	$6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$	2	2
7	$7 = 6 + 1 = 5 + 2 = 4 + 3 = 4 + 2 + 1$	2	3

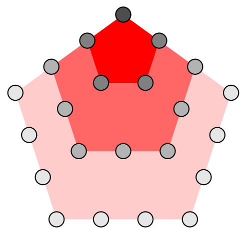
2 Pentagonal number theorem

Theorem 7 (Euler)**.**

$$\prod_{i=1}^{\infty}(1-x^i) \;=\; 1+\sum_{k=1}^{\infty}(-1)^k\,(x^{(3k^2-k)/2}+x^{(3k^2+k)/2}) \;=\; \sum_{k\in\mathbb{Z}}(-1)^k\,x^{(3k^2-k)/2}.$$

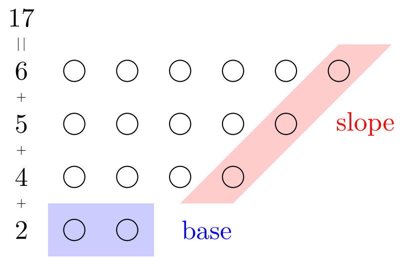
Remark 8 (Pentagonal numbers)**.**

k	*-3*	*-2*	*-1*	*0*	*1*	*2*	*3*
$\frac{3k^2-k}{2}$	*15*	*7*	*2*	*0*	*1*	*5*	*12*



2.1 Proof

Consider a diagram of a partition of n into different parts.



We define the following two operations on the diagrams and denote them α and β.

α **:** If $|b| \le |s|$ and if b and s have no common point, or if $|b| \le |s|-1$, then we move b up and turn its points into new slope.

β **:** If $|b| > |s|$ and if b and s have no common point, or if $|b| \ge |s|+2$, then we move the slope down and turn its points into the new base.

Remark 9. *If b and s have a common point and $|b| = |s|$ or $|b| = |s|+1$ then neither α nor β can be applied.*

α

β

To each diagram D we can apply at most one of the operations α or β. If α can be applied to D then β can be applied to $\alpha(D)$. If β can be applied to D then α can be applied to $\beta(D)$.

Thus we have a 1:1 correspondence

$$\left\{ \begin{array}{c} \text{odd partitions} \\ \text{into distinct parts} \\ \text{s. t. } \alpha \text{ or } \beta \text{ can be applied} \end{array} \right\} \leftrightarrow \left\{ \begin{array}{c} \text{even partitions} \\ \text{into distinct parts} \\ \text{s. t. } \alpha \text{ or } \beta \text{ can be applied} \end{array} \right\}$$

When can neither α nor β be applied?

- b and s have a common point and $|b| = |s| = k$

$$n = k + (k+1) + \ldots + (2k-1) = \frac{3k^2 - k}{2}.$$

- b and s have a common point and $|b| - 1 = |s| = k$

$$n = (k+1) + \ldots + 2k = \frac{3k^2 + k}{2}.$$

Therefore:

$$p_e(n) - p_o(n) = c_n = \begin{cases} (-1)^k, & \text{if } n = \frac{3k^2 \pm k}{2} \\ 0, & \text{otherwise.} \end{cases}$$

This finishes the proof.

3 Jacobi triple product

Theorem 10. *For all $w \neq 0$ and q with $|q| < 1$:*

$$\prod_{k=1}^{\infty} (1 - q^{2k})\,(1 - q^{2k-1}\,w^2)\,(1 - q^{2k-1}\,w^{-2}) = \sum_{k=-\infty}^{\infty} (-1)^k\, w^{2k}\, q^{k^2}.$$

3.1 Proof

Set $\phi_n(w,q) := \prod_{k=1}^{n} (1 - q^{2k-1}\,w^2)\,(1 - q^{2k-1}\,w^{-2})$. We have

$$\phi_n(qw, q) \;=\; \phi_n(w,q)\,\frac{(1 - q^{2n+1}\,w^2)\,(1 - q^{-1}w^{-2})}{(1 - qw^2)\,(1 - q^{2n-1}w^{-2})} \;=\; \phi_n(w,q)\,\frac{1 - q^{2n+1}\,w^2}{-qw^2 + q^{2n}}. \tag{1}$$

We consider the following expansion

$$\phi_n(w,q) = \sum_{k=-n}^{n} A_{n,k}(q)\, w^{2k}, \tag{2}$$

where $A_{n,k}(q)$ is a polynomial in q. We have a symmetry $A_{n,k}(q) = A_{n,-k}(q)$. We find

$$A_{n,n}(q) = q^{1+3+5+\ldots+(2n-1)} = q^{n^2}. \tag{3}$$

We substitute (2) into (1) and obtain

$$A_{n,k}(q) = A_{n,k-1}(q)\, q^{2k-1}\, \frac{1-q^{2n-2k+2}}{1-q^{2n+2k}}. \tag{4}$$

From (3) and (4) we find

$$A_{n,k} = \frac{q^{k^2}}{(1-q^2)\cdots(1-q^{2n})} \prod_{s=n-k+1}^{n} (1-q^{2s}) \prod_{s=n+k+1}^{2n} (1-q^{2s}). \tag{5}$$

From (5) we obtain

$$\prod_{k=1}^{n} (1-q^{2k})\,(1-q^{2k-1}\,w^2)\,(1-q^{2k-1}\,w^{-2}) = \sum_{k=-n}^{n} (-1)^k w^{2k}\, q^{k^2}\, B_{n,k}(q).$$

where

$$B_{n,k}(q) = \prod_{s=n-k+1}^{n} (1-q^{2s}) \prod_{s=n+k+1}^{2n} (1-q^{2s}).$$

We have $B_{n,k} = 1 + O(q^{n-k+1})$. We take the limit $n \to \infty$ and finish the proof.